



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER II, SESSION 2024/2025

BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECJ3553 ARTIFICIAL INTELLIGENCE - SECTION 01

PROJECT PROTOTYPE

THEME: HEALTHCARE

(MINDCARE AI - INTELLIGENT MENTAL HEALTH SUPPORT CHATBOT)

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SUBMISSION DATE : 21 JANUARY 2026

PRESENTATION LINK: <https://youtu.be/gn14OeVFJSk>

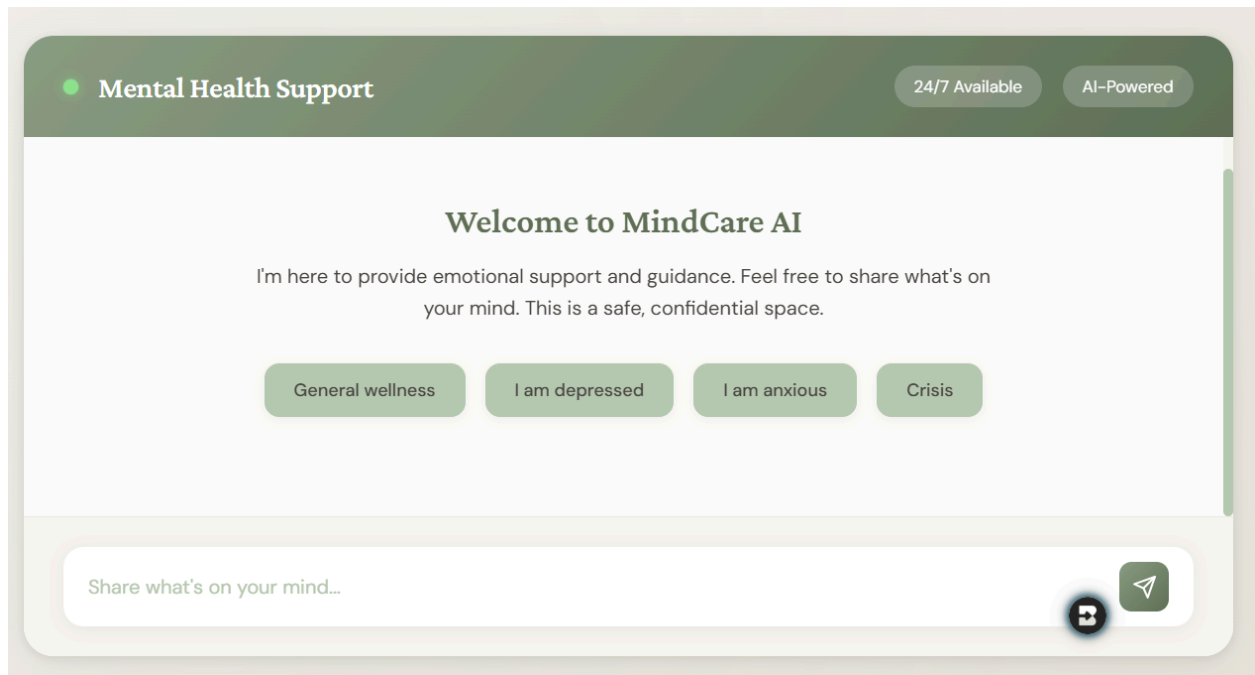
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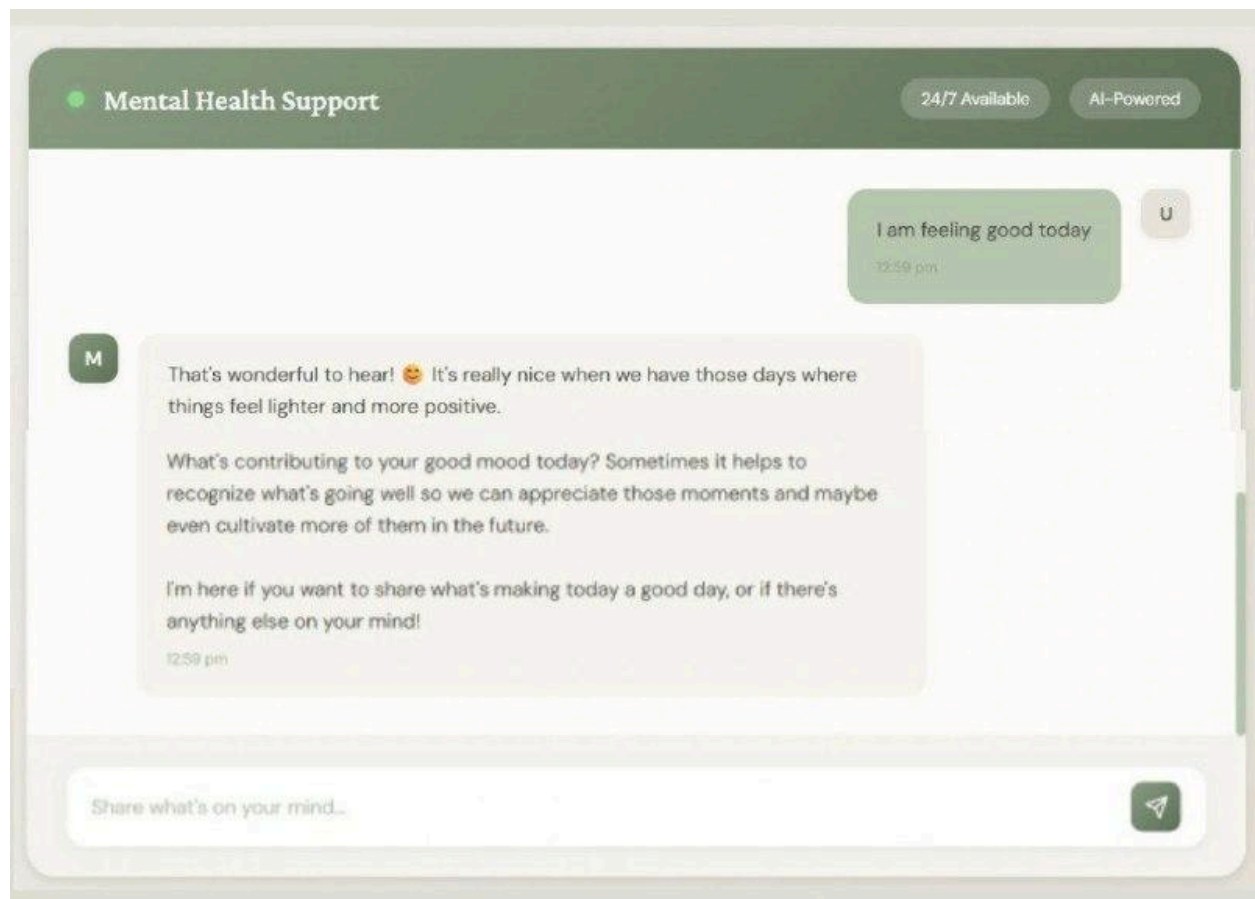
1.0 Overview

This Proof of Concept (POC) demonstrates the practical implementation of MindCare AI, an intelligent mental health support chatbot that integrates the knowledge representation rules from Assignment 1, the state space search logic from Assignment 2, and the intelligent agent behaviors from Assignment 3. The interactive prototype showcases how the system perceives user inputs through natural language processing, applies First-Order Logic rules to identify mental health conditions such as depression and anxiety, and delivers appropriate responses ranging from breathing exercises to emergency crisis intervention. Through this functional demonstration, we validate that our AI solution can accurately detect emotional distress symptoms, prioritize user safety by immediately identifying high-risk situations, and provide empathetic, evidence-based mental health support in real-time, proving the feasibility and effectiveness of our proposed intelligent healthcare system.

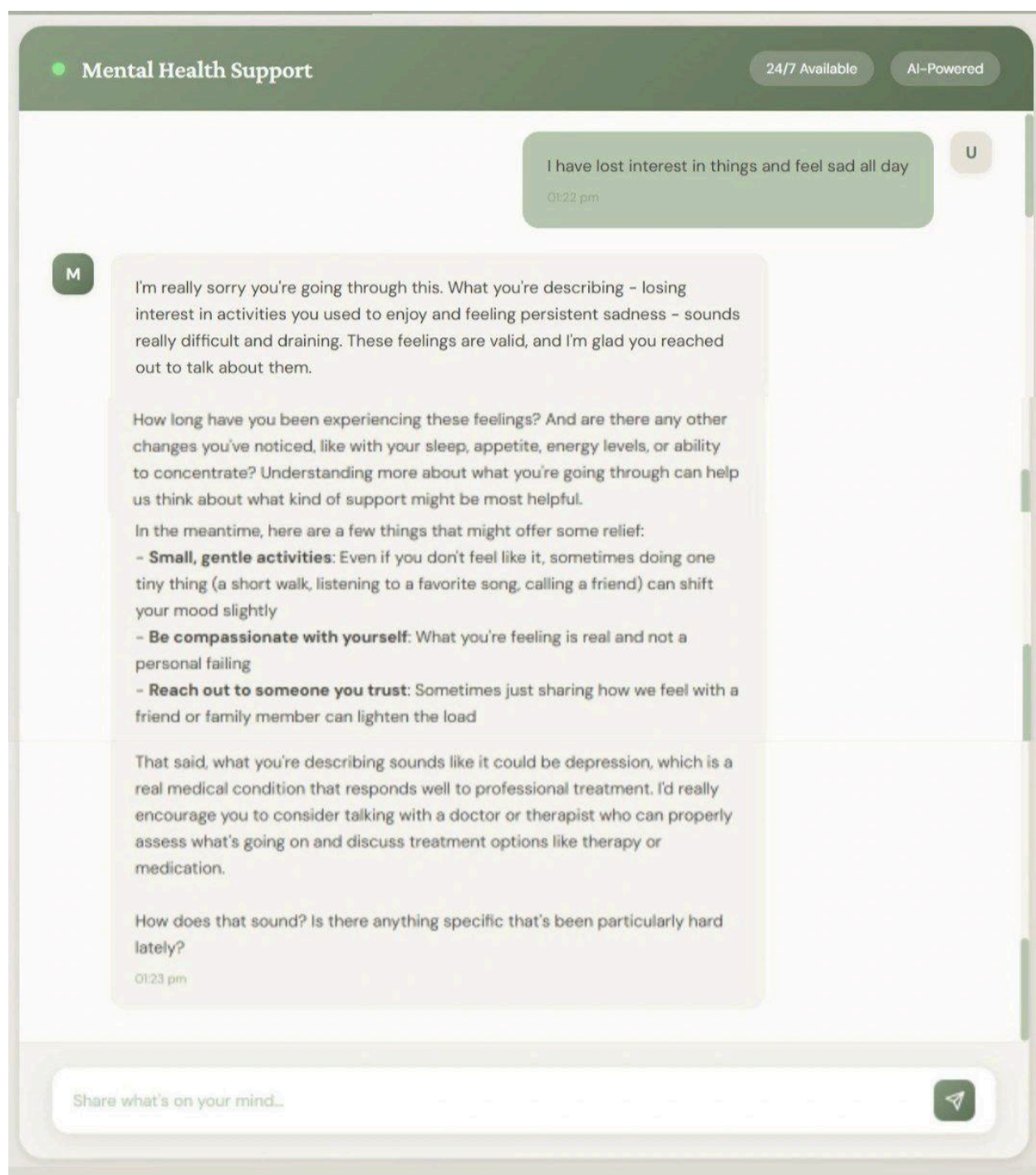
2.0 Proof of Concept (Prototype)



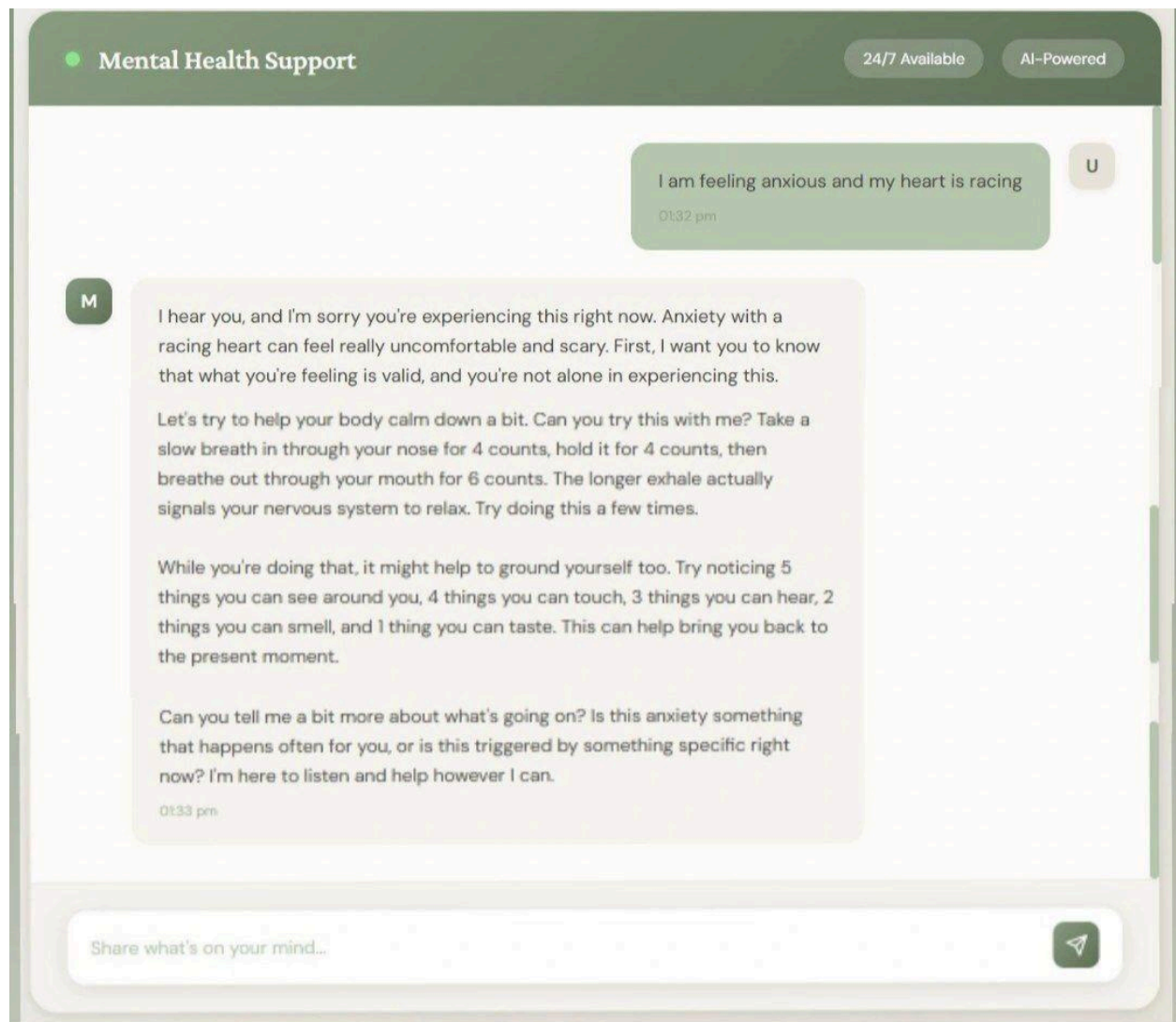
This figure shows the MindCare AI main interface, including the header that indicates the system is AI-powered and available 24/7. The welcome message introduces the purpose of the platform, highlighting that it is a safe and confidential space. The four quick action buttons (*General wellness*, *I am depressed*, *I am anxious*, *Crisis*) are clearly displayed, allowing users to start a conversation easily. Each button pre-fills a message that the user can further edit or expand, lowering the barrier to initiating a conversation and allowing users to express their emotions with minimal effort.



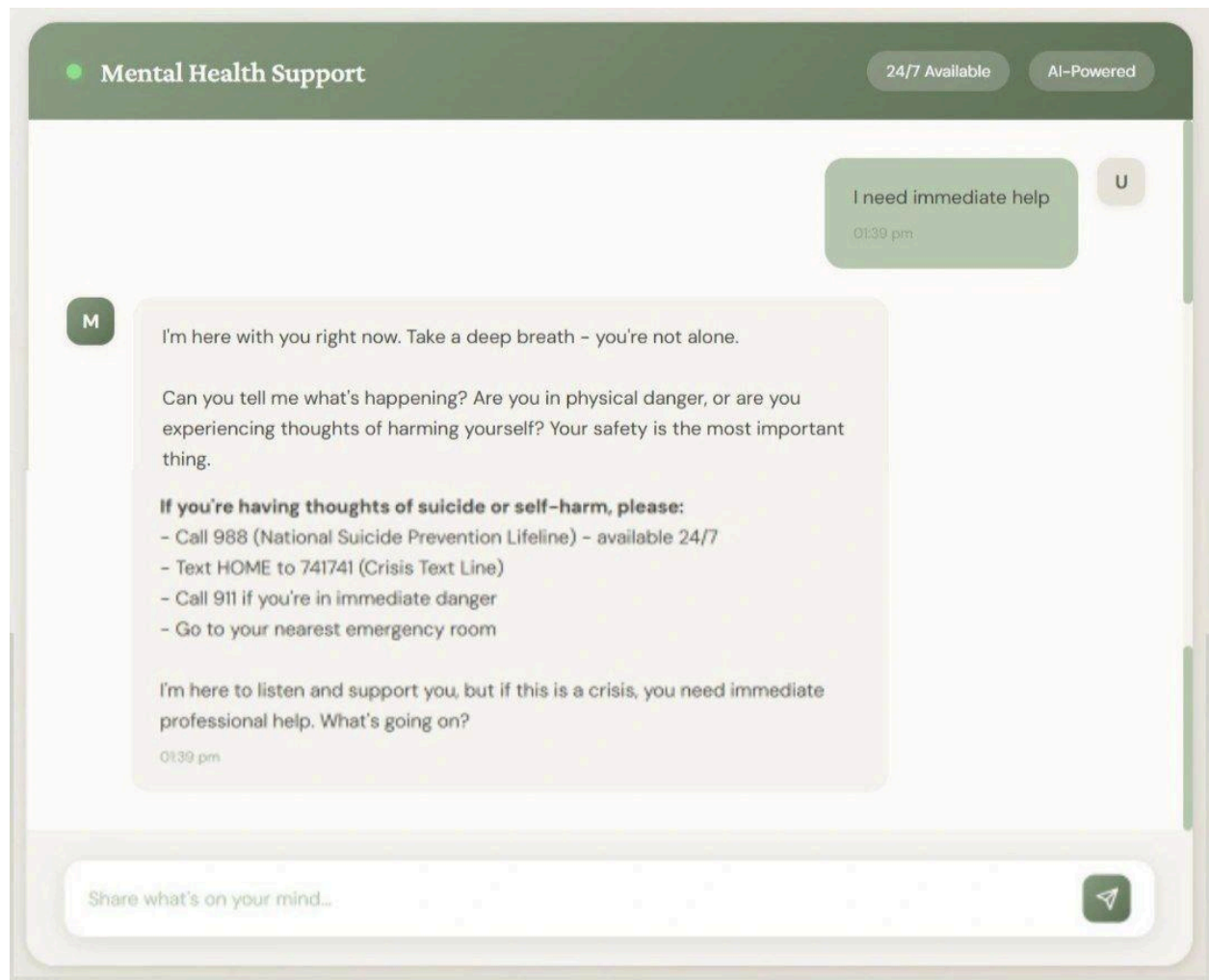
This figure illustrates the AI response when a user selects the *General wellness* button. The prefilled message “I am feeling good today” prompts the AI to provide a positive, validating response that encourages emotional reflection and self-awareness.



This figure demonstrates the AI response after selecting *I am depressed*. The AI responds empathetically, validates the user's emotions, and provides coping strategies while encouraging professional mental health support. This interaction highlights the system's approach to sensitive emotional situations.



This figure shows the AI guiding the user through a 4-7-8 breathing technique when the *I am anxious* option is selected. The interface displays step-by-step instructions and supportive messages, helping the user manage immediate anxiety symptoms in a safe and calming manner.



This figure illustrates the AI's crisis mode. When the user selects *Crisis*, the AI provides emergency contact information and helpline numbers, emphasizes the importance of immediate professional support, and maintains an empathetic and supportive tone. This ensures user safety while responding to high-risk situations.